

Point Class Documentation

Source File: 'Point.h'

Namespace: dsp

Class Header: class Point : public Object

Overview

The *Point* class is a bounded two-dimensional point.

Constructors

- `Point()` (default constructor)
 - **Purpose:** Constructs a point in the upper left-hand corner of a 3 by 3 boundary.
- `Point(size_t r, size_t c)`
 - **Purpose:** Constructs a point in the upper left-hand corner of a *r* by *c* boundary only if both parameters are not zero.
 - **Parameter(s):**
 - *r*: Unsigned integer.
 - *c*: Unsigned integer.
 - **Exception(s):**
 - *invalid_argument*: Occurs if either *r* or *c* is 0.
- `Point(const Point& obj)` (copy constructor)
 - **Purpose:** Makes a deep copy of *obj*.
 - **Parameter(s):**
 - *obj*: Constant *Point* reference object.

Assignment Operators

- `operator=(const Point& rhs)`
 - **Purpose:** Makes a deep copy of *rhs*.
 - **Parameter(s):**
 - *rhs*: Constant *Point* reference object.
 - **Return:** **this*.

Destructor

- `~Point()` [virtual]
 - **Purpose:** Does nothing

Methods

- `rows() const`
 - **Purpose:** Retrieves the number of rows in its boundary.
 - **Return:** An unsigned integer.
- `columns() const`
 - **Purpose:** Retrieves the number of columns in its boundary.
 - **Return:** An unsigned integer.
- `rows(size_t obj)`
 - **Purpose:** Changes the number of rows of its boundary to *obj* and reposition itself to the upper left-hand corner only if *obj* is greater than 0.
 - **Parameter(s):**
 - *obj*: An unsigned integer.

- `columns(size_t obj)`
 - **Purpose:** Changes the number of columns of its boundary to *obj* and reposition itself to the upper left-hand corner only if *obj* is greater than 0.
 - **Parameter(s):**
 - *obj*: An unsigned integer.
- `row() const`
 - **Purpose:** Retrieves its row position in its boundary.
 - **Return:** An unsigned integer.
- `column() const`
 - **Purpose:** Retrieves its column position in its boundary.
 - **Return:** An unsigned integer.
- `row(size_t obj)`
 - **Purpose:** Changes its row position to *obj* only if *obj* is less than the number of rows of its boundary.
 - **Parameter(s):**
 - *obj*: An unsigned integer.
- `column(size_t obj)`
 - **Purpose:** Changes its column position to *obj* only if *obj* is less than the number of columns of its boundary.
 - **Parameter(s):**
 - *obj*: An unsigned integer.
- `set(size_t r, size_t c)`
 - **Purpose:** Changes its row and column positions to *r* and *c*, respectively, only if *r* and *c* are within its boundary.
 - **Parameter(s):**
 - *r*: An unsigned integer.
 - *c*: An unsigned integer.
- `hash() const`
 - **Purpose:** Retrieves the one-dimensional representation of the point.
 - **Return:** An unsigned integer.
- `left() const`
 - **Purpose:** Returns a point a column left of the point if the point can be moved left; otherwise, it returns a copy of the point.
 - **Return:** A *Point* object.
- `right() const`
 - **Purpose:** Returns a point a column right of the point if the point can be moved right; otherwise, it returns a copy of the point.
 - **Return:** A *Point* object.
- `up() const`
 - **Purpose:** Returns a point a row above the point if the point can be moved up; otherwise, it returns a copy of the point.
 - **Return:** A *Point* object.
- `down() const`
 - **Purpose:** Returns a point a row below the point if the point can be moved down; otherwise, it returns a copy of the point.
 - **Return:** A *Point* object.
- `toString() const` [overridden]
 - **Purpose:** Returns a string in the format: (*row*, *column*).
 - **Return:** A string.

Mon-Member Function

- `equal(const Point& lhs, const Point& rhs)`
 - **Purpose:** Returns true if *lhs* and *rhs* are in the same boundary.
 - **Parameter(s):**
 - *lhs*: Constant *Point* reference object.
 - *rhs*: Constant *Point* reference object.
 - **Return:** A Boolean.
- `operator==(const Point& lhs, const Point& rhs)`
 - **Purpose:** Returns true if *lhs* and *rhs* are at the same position in the same boundary.
 - **Parameter(s):**
 - *lhs*: Constant *Point* reference object.
 - *rhs*: Constant *Point* reference object.
 - **Return:** A Boolean.
- `operator!=(const Point& lhs, const Point& rhs)`
 - **Purpose:** Returns true if *lhs* and *rhs* are either not at the same position or not in the same boundary.
 - **Parameter(s):**
 - *lhs*: Constant *Point* reference object.
 - *rhs*: Constant *Point* reference object.
 - **Return:** A Boolean.